1. **Identify missing values**

- **Finding:** Only horsepower has missing values, **6 rows (1.51%)**.
- **Cause:** Missing values were coded as "?" and needed conversion to NaN before analysis.
- **Impact:** Any analysis involving horsepower needs either dropping those rows or imputing horsepower.

```
[6]
print(pd.DataFrame({"missing": missing, "missing_%": missing_pct}).sort_values("missing", ascending=False))
```

```
...
               missing  missing_%
horsepower           6       1.51
mpg                  0       0.00
cylinders            0       0.00
displacement         0       0.00
weight               0       0.00
acceleration         0       0.00
model year           0       0.00
origin               0       0.00
car name             0       0.00
```

2. **Estimate skewness and kurtosis**

- **Skewness (highest):** horsepower is the most right-skewed, **skew ≈ 1.09**.
- **Other skew signals:** origin is also skewed (**≈ 0.92**), but this is because it is an encoded category, not a true continuous variable.
- **Kurtosis:** Several variables have **negative kurtosis** (flatter tails than a normal distribution), for example cylinders and model year (your output shows negative values). This supports treating distributions as non-normal rather than assuming Gaussian shape.

```
[7]
num = df.select_dtypes(include=[np.number])
```

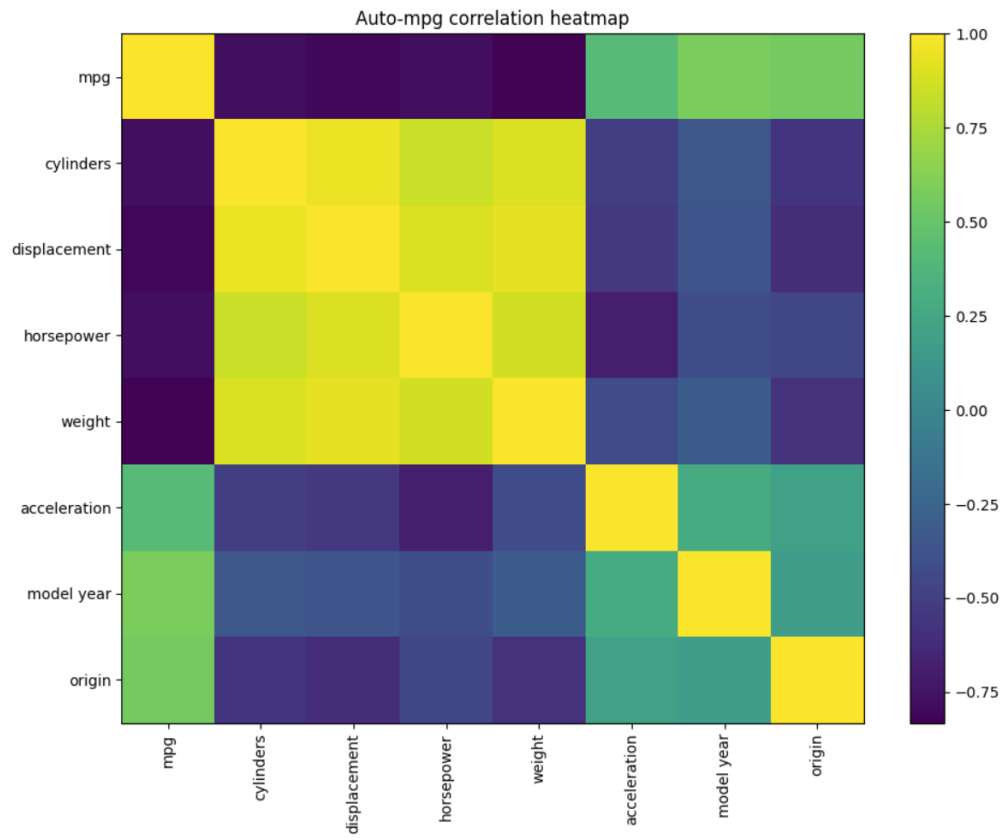
```
[8]
sk = num.skew(numeric_only=True)
ku = num.kurtosis(numeric_only=True)
print(pd.DataFrame({"skewness": sk, "kurtosis": ku}).sort_values("skewness", ascending=False))
```

```
...
               skewness  kurtosis
horsepower     1.087326  0.696947
origin         0.923776 -0.817597
displacement   0.719645 -0.746597
weight         0.531063 -0.785529
cylinders      0.526922 -1.376662
mpg            0.457066 -0.510781
acceleration   0.278777  0.419497
model year     0.011535 -1.181232
```

3. **Correlation heat map**

- **Main pattern:** A strong positive correlation block between **cylinders, displacement, horsepower, weight**.
- **mpg relationship:** mpg is strongly **negatively related** to that block, meaning higher engine size and mass associates with lower fuel efficiency.
- **Caveat:** origin is categorical encoded as numbers, so its correlations are not strictly interpretable as linear numeric relationships.

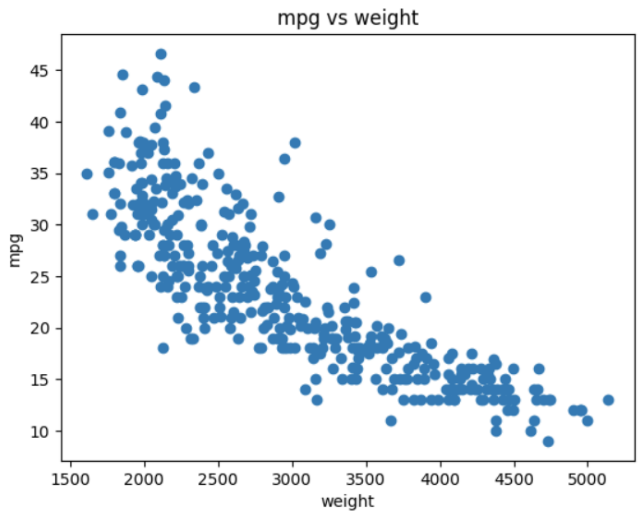
```
corr = num.corr()
plt.figure(figsize=(10, 8))
plt.imshow(corr, aspect="auto")
plt.xticks(range(len(corr.columns)), corr.columns, rotation=90)
plt.yticks(range(len(corr.columns)), corr.columns)
plt.colorbar()
plt.title("Auto-mpg correlation heatmap")
plt.tight_layout()
plt.show()
```

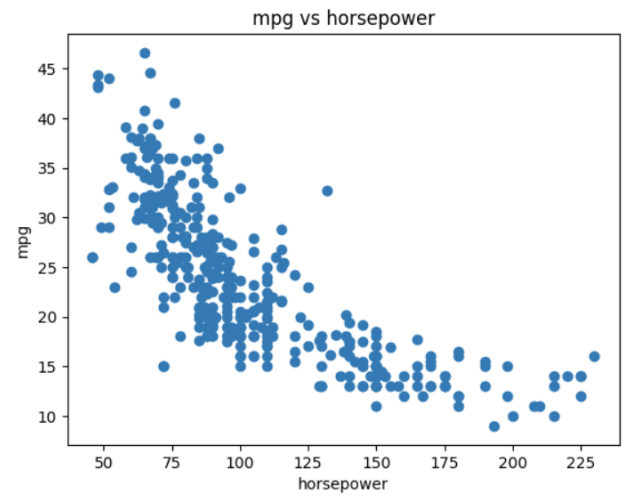


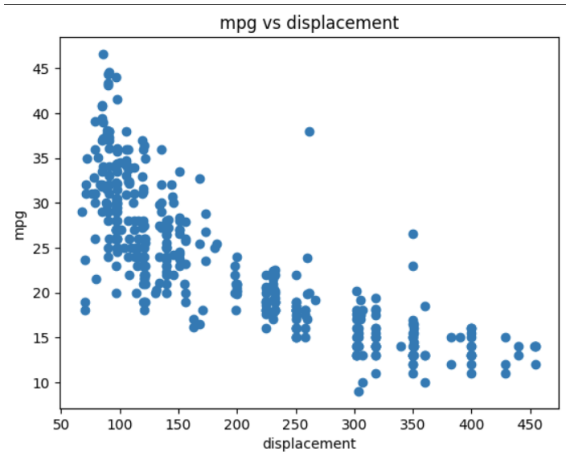
4. **Scatter plot for different parameters**

- **mpg vs weight:** Clear negative trend, heavier cars tend to have lower mpg.
- **mpg vs horsepower:** Clear negative trend, higher horsepower tends to coincide with lower mpg.
- **mpg vs displacement:** Negative trend with visible "banding", consistent with repeated displacement values across models.
- **Interpretation:** Relationships look roughly monotonic with some nonlinearity and outliers, which is useful to know before selecting models or transformations.

```
pairs = [("weight", "mpg"), ("horsepower", "mpg"), ("displacement", "mpg")]
for x, y in pairs:
    plt.figure()
    plt.scatter(df[x], df[y])
    plt.xlabel(x); plt.ylabel(y)
    plt.title(f"{y} vs {x}")
    plt.show()
```







5. **Replace categorical values with numerical values**

- **Origin mapping documented: America = 1, Europe = 2, Japan = 3**.
- **Note:** In this dataset origin is already stored as 1–3, but explicitly recording the mapping prevents misinterpretation.

```
origin_map = {1: "America", 2: "Europe", 3: "Japan"}
print(origin_map)
```

```
{1: 'America', 2: 'Europe', 3: 'Japan'}
```

6. **Mentions for the seminar**

I ran an EDA on auto-mpg. The only missingness was in horsepower, 6 rows or 1.51%, caused by the '?' missing marker.

Horsepower was the most right-skewed variable.

The correlation heatmap showed cylinders, displacement, horsepower, and weight are strongly correlated with each other, while mpg is strongly negatively related to that cluster.

Scatter plots confirmed mpg drops as weight, horsepower, and displacement increase, with some structure and outliers.

I documented the origin encoding as America 1, Europe 2, Japan 3.